

Solve Problems with Unit Rates

Lesson 3-8

Name: Type your name

Date: Today's date

Class: Class period

My Notes

Level 3 Enrichment



TODAY'S OBJECTIVES

Content Objective: I can solve real-world problems by using unit rates to compare options.

Language Objective: I can explain my comparison using the words unit rate, better buy, per unit, and rate.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Unit Rate Problem Solving

Unit Rate

Better Buy

Comparison

Per Unit

Rate

Proportion



Tap any word to see what it means and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

Find the amount per unit, then use it to compare the choices.

2

A rate that gives the amount for exactly one unit is a

.

3

The option that costs less per item, found by comparing unit rates, is the

.

4

Showing how two amounts relate to each other is a

.

5 The cost or amount for a single item is the cost .

6 A ratio comparing two quantities with different units is a .

7 An equation stating that two ratios are equal is a .

Write About the Math

Guided ESOL writing

Use the support level you need. Every level answers the same math question.

1. Understand the Question

EXPLAIN

Which printer should the manager choose, and how long will it take to print all 600 tickets?

Your job: Answer the question. Use math evidence. Explain how the evidence proves your answer.

Responde la pregunta. Usa evidencia matemática. Explica cómo la evidencia demuestra tu respuesta.

2. Plan Your Math Words

Check at least two words you will use.

☐ **Unit Rate Problem Solving** — Find an amount per 1 unit and use it to compare choices or solve a problem.
Resolución de problemas con tasas unitarias — Hallar una cantidad por 1 unidad y usarla para comparar opciones o resolver un problema.

☐ **Unit Rate** — A rate for just 1 of something, like cost for 1 item.
Tasa unitaria — Una tasa para solo 1 de algo, como el precio de 1 artículo.

☐ **Better Buy** — The choice that costs less for each item.
Mejor compra — La opción que cuesta menos por cada artículo.

☐ **Comparison** — Looking at two or more amounts to see how they are related.
Comparación — Mirar dos o más cantidades para ver cómo se relacionan.

☐ **Per Unit** — For each single item.
Por unidad — Por cada artículo.

Say it first: Say your idea to a partner or quietly to yourself before you write.

Di tu idea a un compañero o en voz baja antes de escribir.

Printer A prints ____ tickets per minute. Printer B prints ____ tickets per minute. The manager should use Printer ____ because it is faster. It will take ____ minutes to print 600 tickets.

Di tu respuesta primero: Mi respuesta es ____ porque ____.

3. Build Your Explanation

Model the parts: This model shows the parts of an explanation. It does not answer your writing question.

Claim: Finding the cost per one token lets me compare deals fairly even when the package sizes differ.

Evidence: Students explain the packages have different quantities, so the per-unit cost (unit rate) is needed to compare deals fairly.

Reasoning: This evidence connects the definition of unit rate to the mathematical claim.

Start

Most language support

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **Printer A prints ____ tickets per minute. Printer B prints ____ tickets per minute. The manager should use Printer ____ because it is faster. It will take ____ minutes to print 600 tickets.**

- **My answer is ____.**

Mi respuesta es ____.

Build

Some language support

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is ____.**

Mi afirmación es ____.

- **I know because ____.**

Lo sé porque ____.

- **So ____.**

Entonces ____.

Explain

Light language support

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

- **My claim is ____.**
Mi afirmación es ____.
- **My math evidence is ____.**
Mi evidencia matemática es ____.
- **This proves ____ because ____.**
Esto demuestra ____ porque ____.

4. Check Your Explanation

- ☐ I answered the question.
Respondí la pregunta.
- ☐ I used math evidence: numbers, an equation, a model, or a comparison.
Usé evidencia matemática: números, una ecuación, un modelo o una comparación.
- ☐ I explained how the evidence proves my answer.
Explicué cómo la evidencia demuestra mi respuesta.
- ☐ I used at least two math words.
Usé por lo menos dos palabras matemáticas.
- ☐ I reread complete sentences and punctuation.
Volví a leer mis oraciones completas y la puntuación.

